THE LEAN CANVAS — PHISHGUARD//NEXUS

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Live Platform: https://phishguard-nexus.vercel.app/ (tinyurl.com/pg-nexus) | **Repository:** github.com/rootbysahil/phishguard-nexus

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<td>1. PROBLEM
• Zero-Day Phishing Lifespan: 70%+ of attacks survive <4 hours before mutating, evading static blacklists.
• Black-Box Neural Networks: Modern models output unexplainable probabilities, creating user distrust.
• Privacy & IP Leakage: Scanners performing live DNS lookups leak victim IP.
• Lack of Indian Cybercrime Escalation: No localized recourse for UPI/KYC victims.

Existing Alternatives:
• Google Safe Browsing (lagged 12-48h)
• VirusTotal (slow, leaks IP)
• Heuristic browser popups</td>
<td>4. SOLUTION
• 24 Pure-Python Features: 0 external network queries; <15ms latency.
• Calibrated ML Champion: Logistic Regression with Platt scaling (ROC-AUC: 1.000, F2: 0.9989).
• Directional SHAP XAI: Transparent local feature contribution weights.
• IBM watsonx.ai Advisories: Plain-language guidance in EN, HI, PA.
• Automated Forensic PDF: Real-time audit reports with SHA-256 evidence hashing.
• Native 1930 Integration: 1-click cybercrime.gov.in incident escalation.

8. KEY METRICS
• ML Performance: ROC-AUC (1.0000), F2 Score (0.9989), Recall (1.0000).
• Inference Latency: <15ms per target.
• System Health: 100% serverless uptime on Vercel.
• Usage: Total citizen scans resolved, PDF forensic audits generated.</td>
<td>3. UNIQUE VALUE PROP
Core Message:
"Forensic, zero-latency cyber defense that explains WHY a link is dangerous and WHAT TO DO in regional languages."

High-Level Concept:
VirusTotal + IBM Explainable AI + Indian National 1930 Cybercrime Helpline.

UN SDG Alignment:
• SDG 9 (Target 9.c: Secure digital access)
• SDG 16 (Target 16.4: Reduce cyber fraud)</td>
<td>9. UNFAIR ADVANTAGE
• Zero Data Leakage: GroupShuffleSplit strictly isolated by registered domain.
• Privacy-First Zero-Call Engine: Raw emails/SMS never leave memory; no client IP leakage.
• Indian Cybercrime Co-Pilot: Official national reporting hotline (1930) and portal escalation integrated natively.

5. CHANNELS
• Vercel Cloud App: https://phishguard-nexus.vercel.app/
• Mobile Shortlink: tinyurl.com/pg-nexus + QR Code PWA
• AICTE & IBM SkillsBuild: Student innovation showcases & webinars.
• Browser Extension / API: Direct integration into email & SMS gateways.</td>
<td>2. CUSTOMER SEGMENTS
• Indian Smartphone Users: 800M+ citizens transacting on UPI / Net Banking.
• Elderly & First-Time Internet Users: Vulnerable to fake KYC & lottery scams.
• Indian MSMEs & Startups: Lacking dedicated Security Operations Centers (SOCs).

Early Adopters:
College students, fintech customers, and IT security teams.</td>
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<td colspan="2">7. COST STRUCTURE
• Hosting & Compute: Vercel Serverless (Hobby Tier: $0/month).
• Database: SQLite on-demand storage (<10MB footprint).
• IBM watsonx.ai Token Usage: On-demand inference for Granite LLM advisories.
• Continuous Model Evaluation: Zero-maintenance scikit-learn pipeline.</td>
<td colspan="3">6. REVENUE / SUSTAINABILITY STREAMS
• Free Public Community Tier: Zero-cost defense for Indian citizens and students.
• B2B Enterprise Threat API: Tiered pay-per-scan endpoint for Indian fintechs, banks, and payment gateways.
• Institutional Audits: White-labeled forensic PDF compliance reporting for corporate colleges & SMEs.
• Government Grants: Digital India / Cyber Surakshit Bharat public innovation funding.</td>
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